

Overall Survival - Data imputation

Setup

```
library(tidyverse)
library(here)
library(mice)
library(arrow)
```

Load data

```
data_raw <- read_parquet(here("data", "wide_data_730.parquet"))
```

Format data, add Nelson-Aalen estimates

Everybody had at least 182 days of follow-up.

```
data <- data_raw |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    event_death = if_else(survival_time <= 182, 1, 0),
    survival_time = pmin(survival_time, 182),
    tx = if_else(randomselect_liqplat == "1", 1, 0),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = FALSE)) |>
  select(
    pat_id,
    gender,
    age,
    ecog_fstcnt,
    tx,
    albumin,
    bilirubin,
    c_reactive_protein,
    lactate_dehydrogenase,
    event_death,
    event_next_treatment,
    event_progression,
    n_biopsies,
    n_blood_products_all,
    n_imaging_all,
    n_stay_plus_ER,
    survival_time,
    diagnosis
  ) |>
  relocate(pat_id, event_death, survival_time)

data$na_est <- nelsonaalen(data, timevar = survival_time, statusvar = event_death)
```

Variables for final analysis

When analyzing the trial data, we will use the following predictors for imputation and modeling:

Demographics and baseline clinical:

- age - age at enrollment
- gender - patient sex
- ecog_fstcnt - ECOG performance status (ordered factor: 0-4)
- diagnosis - primary cancer diagnosis (lumped to 5 most frequent categories)

Laboratory values:

- albumin - serum albumin (g/L)
- c_reactive_protein - CRP (mg/L)
- bilirubin - serum bilirubin (μ mol/L)
- lactate_dehydrogenase - LDH (U/L)

Healthcare utilization:

- n_stay_plus_ER - total days in hospital or emergency room
- n_blood_products_all - number of blood product units received

Clinical events:

- event_next_treatment - indicator for starting next line of treatment
- event_progression - indicator for disease progression

Outcome tracking:

- event_death - death indicator
- na_est - Nelson-Aalen cumulative hazard estimate
- tx - treatment group allocation

Multiple imputation

```
#check if imputation methods correct
init <- mice(data, maxit = 0)
meth <- init$method
pred_matrix <- init$predictorMatrix

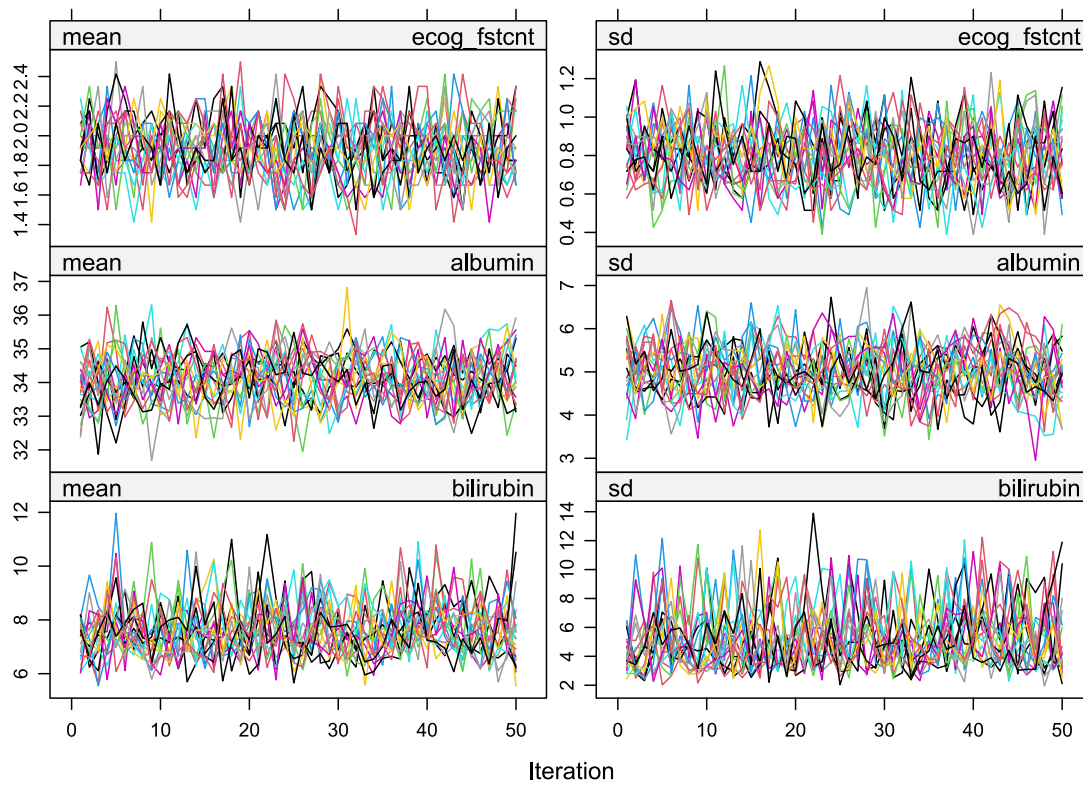
# ensure patient ID and survival time are not used for imputation
pred_matrix[, "survival_time"] <- 0
pred_matrix["survival_time", ] <- 0
pred_matrix[, "pat_id"] <- 0
pred_matrix["pat_id", ] <- 0
meth["pat_id"] <- ""
meth["survival_time"] <- ""

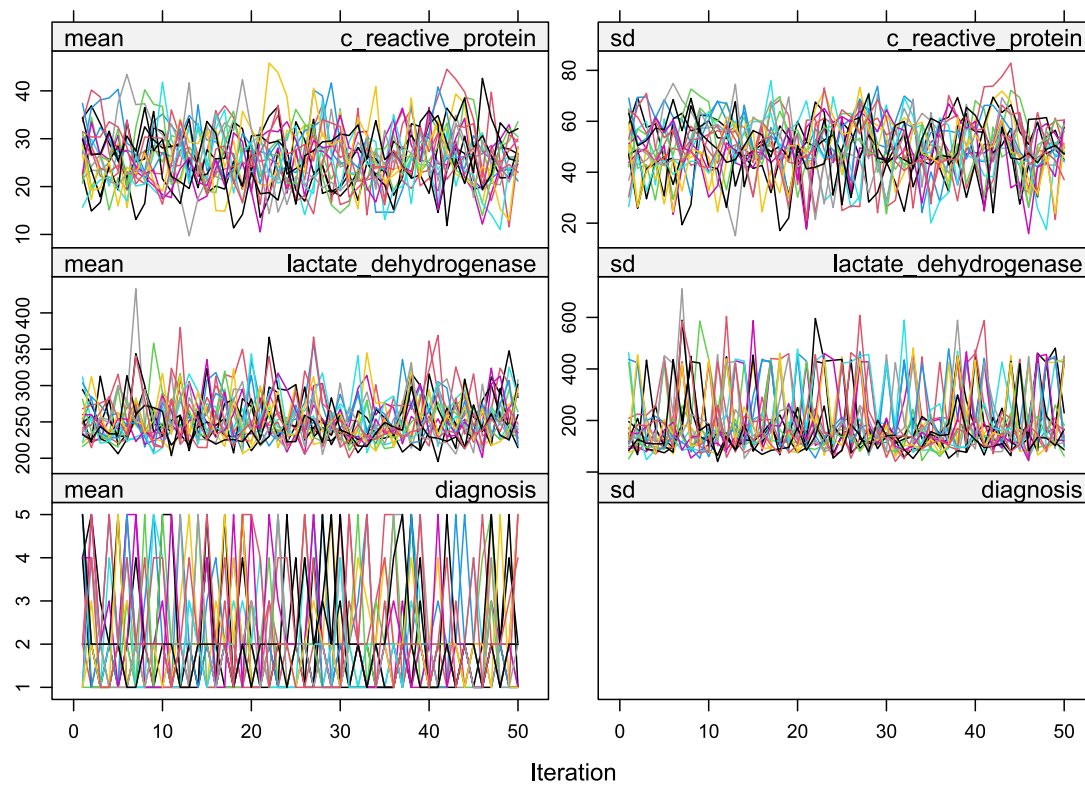
imputed_data <- mice(data,
  m = 20, # final analysis uses 50
  maxit = 50,
  predictorMatrix = pred_matrix,
```

```
method = meth,  
seed = 1234)
```

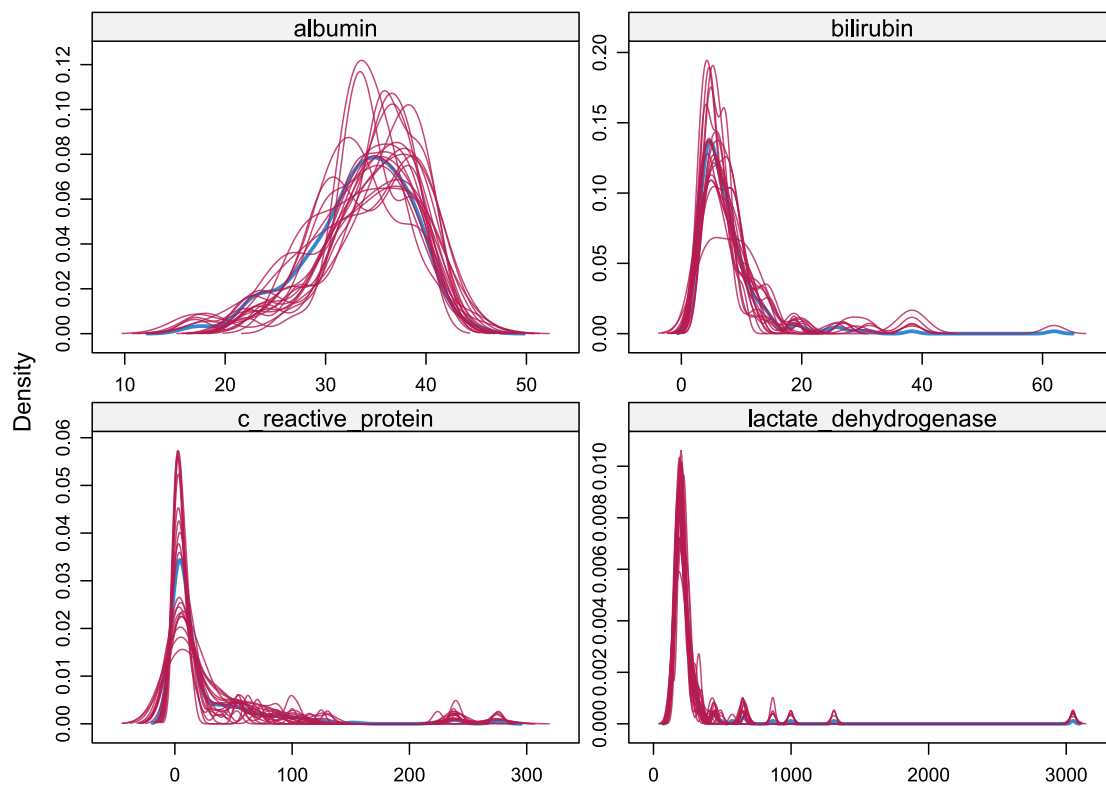
Diagnostics

```
plot(imputed_data)
```

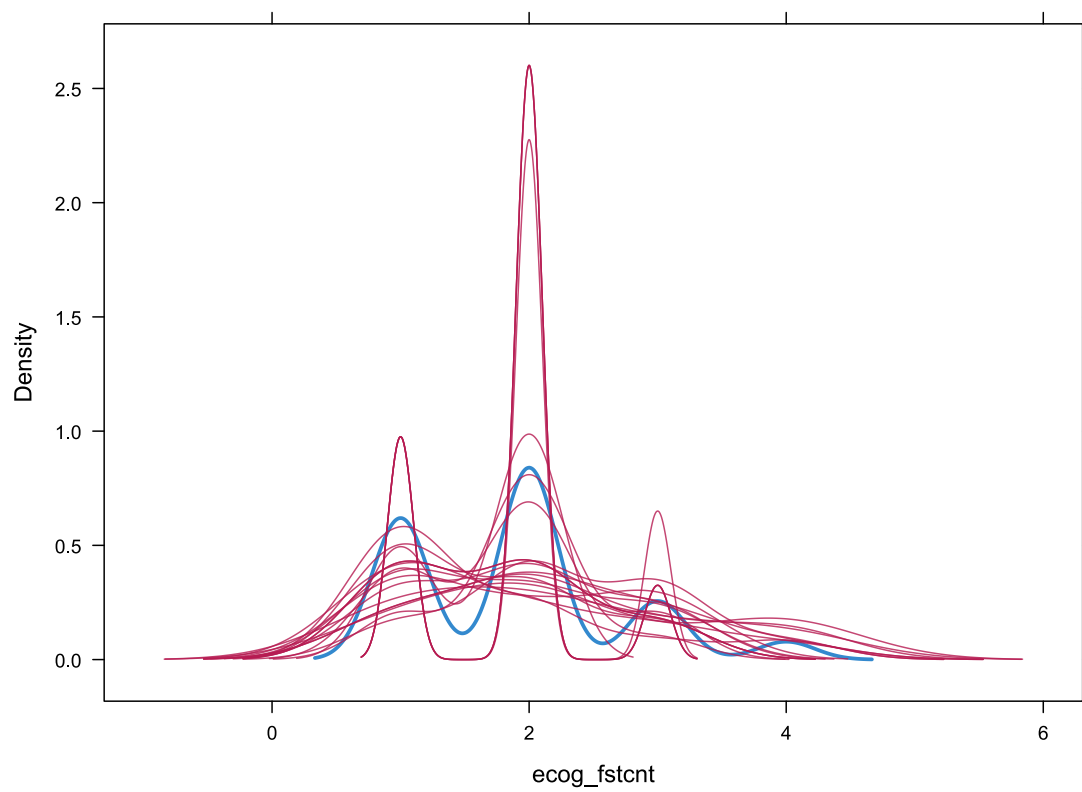




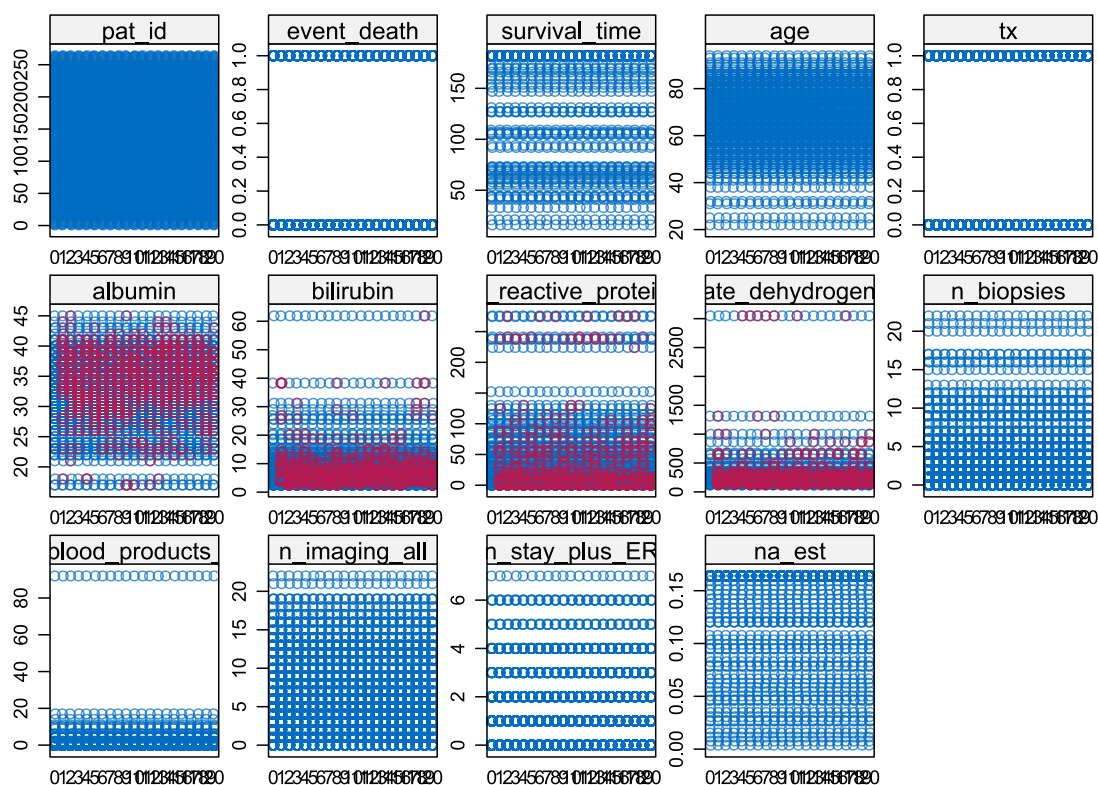
```
densityplot(imputed_data)
```




```
densityplot(imputed_data, ~ecog_fstcnt)
```



```
stripplot(imputed_data)
```



In this data `diagnosis_cat` was only missing in 1 patient.

Select one imputation

For the analysis of the trial data we'd of course use all the imputed datasets and stack the posterior draws. However, for our simulations this is unfeasible, and we shall thus only use one complete dataset for the setup. I will use the data imputed in long format for the time being.

```
complete_data <- complete(imputed_data, 4)
```

Export

```
write_parquet(complete_data, sink = here("data", "survival_complete.parquet"))
```